

MidTerm Exam Monsoon 2025, Big Data Analytics, CSE557

Time: 1 hours

MM:30

All questions are mandatory. Marking is binary. No negative marking.

Roll Number:

Name:

Q1: You as a data scientist got a task of analysing effectiveness of student groups over different courses at IIITD. The dataset you get contains relations $\text{MemberOf}(\text{sid}, \text{gid})$ that captures membership information of a student in different groups, $\text{Task}(\text{gid}, \text{tid})$ that associates each group with the tasks submitted by them, and $\text{Contributes}(\text{sid}, \text{tid}, \text{score})$ that captures the information of actual contribution of a student. It is possible that a student's group has submitted a task but that student had no contribution and hence would have a score of 0. You decide to perform a join of the three relations to get tuples $(\text{sid}, \text{gid}, \text{tid}, \text{score})$. The problem can be mapped to the join of three relations $R(A,B) \bowtie S(B,C) \bowtie T(A,C,D)$ as discussed in the class. You decide to implement the join as a single MapReduce process that minimizes the overall communication cost. Assume that there are B Reduce tasks, and the sizes of relations R , S , and T are $2^{20} = 1,048,576$, $2^{17} = 131,072$, and $2^{14} = 16,384$, respectively. Compute the number of buckets a , b , and c into which each of the attributes A , B , and C are to be hashed. Then, determine the number of times each tuple of R , S , and T is replicated by the Map function. Answer the following questions.

- a) [1 Marks] What is the total communication cost in terms of R , S , T , a , b , and c ?

$$\text{Cost} = R + S + T + c * R + a * S + b * T$$

- b) [2 marks] Give an expression to determine the number of buckets c in terms of R , S , T and B .

$$c = \sqrt[3]{\frac{B * S * T}{R^2}}$$

- c) [2 Marks] What is the value of a , b and c if the total number of buckets B are 4096.

$$\mathbf{a=16 \quad b=128 \quad c=2}$$

- d) [2 Marks] How much should B be scaled up so that the bucket counts a , b , and c double?

B should be scaled 8 times to get bucket counts a , b and c double.

- e) [1 Marks] What is the size of the Reducer on average if number of reducers B is 4096?

$$\begin{aligned} \text{Reducer Size } q &= \frac{\text{Communication Cost}}{\text{Number of Reducers}} \\ q &= \frac{c * R + a * S + b * T}{B} = \frac{(2^{21} + 2^{21} + 2^{21})}{2^{12}} = 3 * 2^9 = 1536 \end{aligned}$$

Q2: Consider the incomplete Map Reduce code for computing certain aggregates over the join output result obtained in the first question in the format (**sid, gid, tid, score**). You need to complete the code as asked in comments. Note that the domain of each attribute is same, i.e. there may be a student with ID value 1, gid with value 1 and tid with value 1. Assume availability of a function **frNonZero(L)** that given a list L of pairs returns a fraction of tuples containing non-zero values in the 2nd component.

```
frNonZero([(1,2), (1,0), (1,3)]) = 0.666
```

```
frNonZero ( [(1,0), (1,20)]) = 0.5
```

```
frNonZero ( [(2,0), (2,0)]) = 0.0
```

```
from mrjob.job import MRJob
```

```
class MRStudentAnalytics(MRJob):
```

```
    def mapper(self, _, line):
```

```
        # input: order_id,country,amount
```

```
        try:
```

```
            sid, gid, tid, score = line.strip().split(",")
```

```
            score = float(score)
```

```
        except ValueError:
```

```
            return
```

```
# q2-a. Count total number of tasks    [1+1 Marks]
```

```
    yield __"#TASKS"__, __tid__        # q2-a
```

Key is any unique string and value is tid.

```
# q2-b. Count the fraction of tasks a student scored non-zero scores [1+1+1 Marks]
```

```
    yield __f"ST_Score::{sid}"__, __(tid,score)__ # q2-b
```

Key should have a unique string and sid, value should be a tuple (tid,score).

```
# q2-c. Count total number of tasks each group is associated with [1+1+1 marks]
```

```
    yield __f"GR_Task::{gid}"__, __tid__        # q2-c
```

Key should have a unique string and gid, value should be tid.

```
# q2-d. Count total number of groups    [1+1 marks]
```

```
    yield __"#GROUPS"__, __gid__        # q2-d
```

Key should have a unique string, value should be gid.

```
# q2-e. Count total number of students    [1+1 marks]
```

```
    yield __"#STDS"__, __sid__        # q2-e
```

Key should have a unique string, value should be sid.

```
def reducer(self, key, values):

    if (key == __ "#TASKS"__): # Reducer Code for q2-a
        yield "#TASKS", len(set(list(values)))

    Key should be equal to what was emitted in mapper. Reducer shall yield a
    unique string, the value should be as given in the expression.

    elif (__str(key).find("ST_Score") != -1_): #Reducer Code for task q2-b
        frac = frNonZero(values)

        yield key, frac

    Key should contain the unique string emitted in the mapper. Reducer shall
    yield the key, the value should be computed as a fraction.

    elif (__str(key).find("GR_Task") != -1_): #Reducer Code for task q2-c
        yield key, len(set(list(values)))

    Key should contain the unique string emitted in the mapper. Reducer shall
    yield the key as it is, the value should be computed as a fraction.

    elif (key == __ "#GROUPS"__): #Reducer Code for task q2-d
        yield "#GROUPS", len(set(list(values)))

    Key should be equal to what was emitted in mapper. Reducer shall yield a
    unique string, the value should be as given in the expression.

    else: #Code for task q2-e
        yield "#STDS", len(set(list(values)))

    Key should be equal to what was emitted in mapper. Reducer shall yield a
    unique string, the value should be as given in the expression.
```

Q3: Consider the following matrix:

	C1	C2	C3	C4
R1	0	1	1	0
R2	1	0	1	1
R3	0	1	0	1
R4	0	0	1	1
R5	1	0	1	0
R6	0	1	0	0

Perform a minhashing of the data, with the order of rows: $h_1 = (R_2, R_6, R_1, R_5, R_3, R_4)$, and $h_2 = (R_4, R_6, R_1, R_3, R_5, R_2)$, and $h_3 = (R_2, R_5, R_1, R_6, R_3, R_4)$. Suppose **we use locality-sensitive hashing with one band of three rows**. Assume there are enough buckets available that the hash function for each band can be the identity function (i.e., columns hash to the same bucket if and only if they are identical in the band). Answer the following:

- a) Find the signature of column C2, i.e. hash value for each h1, h2 and h3. [2 Marks]

$$C2 = [R6, R6, R1]$$

- b) Find all the candidate pairs, if any. If there are no candidate pairs write **None**. [2 Marks]

$$C1=[R2, R5, R2] \quad C2 = [R6, R6, R1] \quad C3 =[R2, R4, R2] \quad C4=[R2, R4, R2]$$

$$\text{Candidate Pairs} = (C3, C4)$$

Q4: Consider the following three vectors u, v, w in a 6-dimensional space:

$$u = [4, 1, 0, 0, 2, 0]$$

$$v = [3, 0, 0, 1, 2, 0]$$

$$w = [0, 1, 3, 0, 0, 1]$$

Suppose we construct 3-bit sketches of the vectors by the random hyperplane method, using the randomly generated normal vectors r1, r2, and r3, in that order:

$$r1 = [1, -1, 1, -1, 1, -1]$$

$$r2 = [-1, -1, 1, 1, -1, 1]$$

$$r3 = [1, 1, 1, 1, 1, 1]$$

- a) Construct the sketches of the three vectors u, v, w. [1.5 Marks]

$$u = [1, -1, 1]$$

$$v = [1, -1, 1]$$

$$w = [1, 1, 1]$$

- b) Estimate the pairwise cosine similarities estimates of u, v, and w from their 3-bit sketches. [1.5 Marks]

$$(u,v) = 1$$

$$(u,w) = 2/3$$

$$(v,w) = 2/3$$

- c) Using Euclidean LSH with bucket length a=1 and three projection lines from the origin to r1, r2, r3 compute the signature of each object u,v,w given

$$r1=[1,-1,1,-1], \quad r2=[-1,-1,1,1], \quad r3=[1,1,1,1] \text{ and}$$

$$u=[4,1,2,0], \quad v=[3,0,0,1], \quad w=[0,0,0,1]$$

And compute the candidate pairs. A pair of points is called a candidate pair if they share a hash value for any line. Buckets are $[-k, -k+1)=-k, \dots, [0, 1)=0, [1, 2)=1, \dots, [n, n+1)=n$

[1 + 2 Marks]

$$\text{Signature of } u? \quad u = [2, -2, 3]$$

Candidate Pairs: As the sketches for points are $u = [2, -2, 3]$, $v=[1,-1, 2]$, $w=[-1, 0, 0]$
There is no hash value common for any function, hence there is no candidate pair.